

Task 2:

Using the MSE loss
$$L(\beta) = \sum_{i=1}^N (y_i^* - x_i \beta)^2$$
$$= (Y^* - X\beta)^T (Y^* - X\beta)$$

and its derivative (with $y_i^* \in \{-1, 1\}$)

$$\begin{aligned} \frac{\partial L}{\partial \beta} &= \frac{\partial}{\partial \beta} (Y^* - X\beta)^T (Y^* - X\beta) \\ &= \frac{\partial}{\partial \beta} (Y^*)^T Y^* - 2 \frac{\partial}{\partial \beta} (Y^*)^T X\beta + \frac{\partial}{\partial \beta} (X\beta)^T (X\beta) \\ &= -2X^T Y^* + 2X^T X\beta \end{aligned}$$

Setting $\frac{\partial L}{\partial \beta} \stackrel{!}{=} 0$ yields

$$\boxed{X^T Y^* = X^T X \beta} \quad (4)$$

Now using $X_i = (X_i - \mu_c) + \mu_c$ for class $c \in \{-1, 1\}$

$$\Rightarrow X^T X = \sum_{i=1}^N ((X_i - \mu_c) + \mu_c)^T ((X_i - \mu_c) + \mu_c)$$

$$= \sum_{i: y_i^* = -1} ((X_i - \mu_{-1}) + \mu_{-1})^T ((X_i - \mu_{-1}) + \mu_{-1})$$

$$+ \sum_{i: y_i^* = 1} ((X_i - \mu_1) + \mu_1)^T ((X_i - \mu_1) + \mu_1)$$

$$= \sum_{i: y_i^* = -1} (X_i - \mu_{-1})^T (X_i - \mu_{-1}) + \sum_{i: y_i^* = -1} \underbrace{(X_i - \mu_{-1})^T \mu_{-1}}_{=0}$$

$$+ \mu_{-1}^T \underbrace{\sum_{i: y_i^* = -1} X_i - \mu_{-1}}_{=0} + N_{-1} \mu_{-1}^T \mu_{-1}$$

$$+ \sum_{i: y_i^* = 1} (X_i - \mu_1)^T (X_i - \mu_1) + \sum_{i: y_i^* = 1} \underbrace{(X_i - \mu_1)^T \mu_1}_{=0}$$

$$+ \mu_1^T \underbrace{\sum_{i: y_i^* = 1} X_i - \mu_1}_{=0} + N_1 \mu_1^T \mu_1$$

$$= \sum_{i: y_i^* = -1} (X_i - \mu_{-1})^T (X_i - \mu_{-1}) + \sum_{i: y_i^* = 1} (X_i - \mu_1)^T (X_i - \mu_1) + N_{-1} \mu_{-1}^T \mu_{-1} + N_1 \mu_1^T \mu_1$$

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$$\Rightarrow X^T X = N\Sigma + N_{-1}\mu_{-1}^T\mu_{-1} + N_1\mu_1^T\mu_1$$

Since $N_{-1} = N_1 = \frac{N}{2}$ and $\mu_{-1} + \mu_1 = 0 \Rightarrow \mu_{-1} = -\mu_1$

$$\Rightarrow X^T X = N\Sigma + \frac{N}{2}((- \mu_1)^T(-\mu_1) + \mu_1^T\mu_1)$$

$$= N\Sigma + N\mu_1^T\mu_1$$

Again, as $\mu_{-1} = -\mu_1 \Rightarrow \mu_{-1}^T\mu_{-1} = 2\mu_1^T\mu_1$, so

$$\mu_1^T\mu_1 = \frac{1}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1})$$

$$\Rightarrow X^T X = N\Sigma + \frac{N}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1}) \quad (44)$$

Substituting (44) into (4) together with

$$X^T Y^* = \sum_{i: Y_i^* = 1} X_i^T - \sum_{i: Y_i^* = -1} X_i^T$$

$$= N_1\mu_1^T - N_{-1}\mu_{-1}^T$$

$$= \frac{N}{2}(\mu_1^T - \mu_{-1}^T)$$

$$\Rightarrow X^T Y^* = X^T X \beta \Leftrightarrow (N\Sigma + \frac{N}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1}))\beta = \frac{N}{2}(\mu_1 - \mu_{-1})^T$$

$$\Leftrightarrow \left[\Sigma\beta + \frac{1}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1})\beta = \frac{1}{2}(\mu_1 - \mu_{-1})^T \right]$$

Let $\tau' := (\mu_1 - \mu_{-1})\beta$, so

$$\Sigma\beta + \frac{1}{4}(\mu_1 - \mu_{-1})^T\tau' = \frac{1}{2}(\mu_1 - \mu_{-1})^T$$

$$\Leftrightarrow \Sigma\beta = \frac{1}{2}(\mu_1 - \mu_{-1})^T - \frac{1}{4}\tau'(\mu_1 - \mu_{-1})^T$$

$$= \left(\frac{1}{2} - \frac{\tau'}{4}\right)(\mu_1 - \mu_{-1})^T$$

Finally, let $\tau := \left(\frac{1}{2} - \frac{\tau'}{4}\right)$ to get

$$\beta = \tau \Sigma^{-1}(\mu_1 - \mu_{-1})^T$$